

## CRESSIER, Switzerland

WMO: 066060

Lat: 47.0475N

Lon: 7.0592E

Elev: 431

StdP: 96.25

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -7.2       | -5.0 | -12.6                         | 1.3 | -7.2 | -8.2 | 2.0 | -3.5 | 11.5                  | 4.1  | 10.3 | 1.9  | 4.3                   | 50   | 0.488 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.1                   | 30.5            | 20.7 | 28.8 | 20.1 | 27.0 | 19.1 | 21.7                | 28.4 | 20.7 | 27.1 | 19.9 | 25.5 | 2.9                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.7                            | 15.2 | 23.6 | 18.7 | 14.3 | 22.6 | 18.0 | 13.6 | 22.0 | 65.0          | 28.4 | 61.7 | 27.1 | 58.8 | 25.5 | 26.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 9.8           | 8.4  | 7.2 | -8.9                       | 33.1 | 2.5                | 1.4 | -10.7                                              | 34.1 | -12.2      | 34.9 | -13.6      | 35.7 | -15.5      | 36.7 |    |    |
| (5)               |      |     | -9.6                       | 24.6 | 2.6                | 0.9 | -11.5                                              | 25.3 | -13.0      | 25.8 | -14.5      | 26.3 | -16.4      | 27.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.7   | 2.1   | 2.0   | 6.4   | 10.5  | 13.6  | 18.2  | 20.0  | 19.6  | 15.7  | 11.1  | 5.9   | 3.0  |
|                                                                                  | DBStd       | 7.23   | 3.45  | 3.82  | 3.09  | 3.43  | 3.38  | 3.18  | 2.90  | 3.02  | 2.76  | 3.01  | 2.97  | 3.10 |
|                                                                                  | HDD10.0     | 997    | 244   | 224   | 115   | 35    | 7     | 0     | 0     | 0     | 0     | 25    | 128   | 218  |
|                                                                                  | HDD18.3     | 2954   | 502   | 458   | 369   | 236   | 151   | 40    | 17    | 22    | 87    | 225   | 373   | 476  |
|                                                                                  | CDD10.0     | 1263   | 0     | 1     | 4     | 49    | 118   | 247   | 311   | 298   | 172   | 58    | 4     | 0    |
|                                                                                  | CDD18.3     | 181    | 0     | 0     | 0     | 0     | 4     | 37    | 69    | 62    | 9     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 1607   | 0     | 0     | 0     | 6     | 46    | 345   | 615   | 527   | 68    | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 433    | 0     | 0     | 0     | 0     | 3     | 96    | 182   | 148   | 4     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.4    | 2.8   | 2.8   | 2.8   | 2.6   | 2.5   | 2.4   | 2.3   | 1.9   | 2.2   | 1.9   | 2.2   | 2.6  |
| Precipitation                                                                    | PrecAvg     | 1211   | 89    | 86    | 80    | 91    | 111   | 108   | 115   | 116   | 100   | 103   | 103   | 110  |
|                                                                                  | PrecMax     | 1569   | 227   | 205   | 277   | 179   | 207   | 220   | 242   | 274   | 182   | 184   | 251   | 237  |
|                                                                                  | PrecMin     | 871    | 21    | 26    | 22    | 21    | 50    | 59    | 50    | 22    | 39    | 18    | 11    | 2    |
|                                                                                  | PrecStd     | 177    | 54    | 45    | 58    | 49    | 45    | 43    | 42    | 52    | 42    | 43    | 58    | 55   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 11.4  | 14.5  | 18.8  | 24.2  | 26.7  | 31.7  | 32.8  | 31.9  | 27.4  | 21.2  | 16.0  | 11.5 |
|                                                                                  |             | MCWB   | 8.8   | 9.2   | 11.0  | 15.3  | 17.8  | 22.0  | 21.2  | 21.0  | 18.5  | 14.9  | 11.9  | 9.6  |
|                                                                                  | 2%          | DB     | 9.7   | 11.1  | 16.9  | 21.6  | 24.6  | 29.5  | 30.6  | 30.1  | 25.5  | 19.5  | 14.0  | 10.2 |
|                                                                                  |             | MCWB   | 8.0   | 7.6   | 10.3  | 13.6  | 16.4  | 20.9  | 20.2  | 20.8  | 18.5  | 14.8  | 11.6  | 8.8  |
|                                                                                  | 5%          | DB     | 8.1   | 9.0   | 14.8  | 19.8  | 22.6  | 27.2  | 28.7  | 28.4  | 23.5  | 17.7  | 12.2  | 9.3  |
|                                                                                  |             | MCWB   | 6.4   | 6.5   | 9.2   | 12.8  | 15.4  | 19.5  | 19.7  | 20.2  | 17.6  | 14.1  | 10.4  | 8.0  |
|                                                                                  | 10%         | DB     | 6.7   | 7.2   | 12.5  | 17.6  | 20.6  | 25.0  | 26.8  | 26.4  | 21.7  | 16.2  | 10.6  | 7.7  |
|                                                                                  |             | MCWB   | 5.5   | 5.1   | 8.2   | 11.7  | 14.5  | 18.2  | 18.8  | 19.1  | 16.7  | 13.6  | 9.3   | 6.6  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 9.7   | 9.7   | 11.6  | 15.5  | 18.7  | 23.6  | 22.3  | 22.4  | 20.6  | 16.8  | 13.2  | 10.1 |
|                                                                                  |             | MCDB   | 10.6  | 12.6  | 17.4  | 22.2  | 25.0  | 30.5  | 29.7  | 28.6  | 23.8  | 18.7  | 14.8  | 10.6 |
|                                                                                  | 2%          | WB     | 8.2   | 7.9   | 10.5  | 14.0  | 17.1  | 21.6  | 21.2  | 21.5  | 19.3  | 15.6  | 11.9  | 9.2  |
|                                                                                  |             | MCDB   | 9.5   | 10.9  | 15.9  | 20.1  | 22.5  | 28.2  | 28.6  | 28.1  | 23.2  | 17.7  | 13.5  | 10.1 |
|                                                                                  | 5%          | WB     | 6.7   | 6.8   | 9.6   | 12.8  | 15.8  | 20.3  | 20.3  | 20.6  | 18.5  | 14.8  | 10.7  | 8.1  |
|                                                                                  |             | MCDB   | 8.0   | 9.1   | 14.1  | 18.2  | 20.7  | 26.2  | 26.8  | 26.8  | 22.3  | 16.8  | 12.1  | 9.2  |
|                                                                                  | 10%         | WB     | 5.6   | 5.4   | 8.5   | 11.8  | 14.7  | 19.1  | 19.5  | 19.8  | 17.5  | 13.9  | 9.4   | 6.8  |
|                                                                                  |             | MCDB   | 6.6   | 7.1   | 12.1  | 16.2  | 19.0  | 23.9  | 25.2  | 25.2  | 20.9  | 15.8  | 10.3  | 7.5  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.8   | 6.4   | 9.2   | 10.2  | 9.8   | 10.4  | 11.1  | 11.3  | 10.0  | 8.0   | 5.3   | 4.6  |
|                                                                                  |             | MCDBR  | 7.1   | 11.1  | 14.4  | 15.0  | 13.9  | 14.3  | 14.7  | 14.2  | 13.0  | 11.1  | 8.3   | 6.5  |
|                                                                                  |             | MCWBR  | 5.4   | 7.5   | 8.2   | 7.5   | 6.3   | 6.6   | 6.0   | 5.8   | 6.9   | 7.1   | 6.1   | 5.4  |
|                                                                                  | 5% WB       | MCDBR  | 6.8   | 10.2  | 12.9  | 13.2  | 11.8  | 13.5  | 13.1  | 13.2  | 11.8  | 9.3   | 7.7   | 6.1  |
| MCWBR                                                                            |             | 5.4    | 7.3   | 7.6   | 6.9   | 5.6   | 6.6   | 6.6   | 5.7   | 5.8   | 6.8   | 6.5   | 5.9   | 5.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.299  | 0.325 | 0.362 | 0.401 | 0.410 | 0.413 | 0.412 | 0.403 | 0.381 | 0.359 | 0.323 | 0.297 |      |
|                                                                                  | taud        | 2.433  | 2.363 | 2.314 | 2.232 | 2.252 | 2.290 | 2.287 | 2.323 | 2.364 | 2.410 | 2.453 | 2.441 |      |
|                                                                                  | Ebn at Noon | 793    | 838   | 856   | 852   | 855   | 854   | 849   | 841   | 826   | 788   | 753   | 750   |      |
|                                                                                  | Edh at Noon | 68     | 89    | 109   | 130   | 132   | 128   | 127   | 117   | 103   | 85    | 67    | 60    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.21   | 2.05  | 3.24  | 4.35  | 5.02  | 5.94  | 5.75  | 5.01  | 3.86  | 2.45  | 1.34  | 0.98  |      |
|                                                                                  | RadStd      | 0.15   | 0.31  | 0.48  | 0.64  | 0.47  | 0.55  | 0.51  | 0.46  | 0.35  | 0.24  | 0.17  | 0.17  |      |

## Historical Trends

|                               | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                               |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                               |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only             | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (108 neighbors) | N/A   | N/A     | N/A    | +0.73   | +0.40 | +0.30 | N/A         | N/A     | +53     | +17     |

Nomenclature: See separate page